

Shekinah Wise Pita - Complete MIT Application (Minus personal details)

JOBS

1. Ongata Ronkai Secondary School (Physics and Mathematics teacher) - Teaching 567 form 4 students physics and mathematics; form mean score has risen by 40%. Hold evening meetings with girls encouraging them to pursue academic goals & avoid being forced into housewives' roles...July 2022 to present.
2. Generation X solutions LTD (Intern) - Provided research support on Nuclear Power and Energy Agency(NuPEA)&Kenya Electricity Generating Company projects; invented public sensitization model. Engaged in pre-project brainstorming & rollout sessions...May 2022 to July 2022.

ACTIVITIES

1. Danish Robotics Exchange Program Cohort (Chairperson) - Worked with 12 teammates to create 10 AGV prototype robots proficient in thermal vision, flexible sortation & autonomous navigation. Prototype robots were adopted by a Danish high school. Trained subsequent cohorts on the same.
2. NuPEA sponsored Kenya Science and Engineering Fair (Team Leader of the Automation Optimization Project) - Fashioned a prototype drive shaft bicycle incorporating hub gears & Hobson's joints to increase bicycle efficiency; Incorporated in a chain-driven wheelchair.
3. Football team (Central midfielder) - Designed attack and defensive formations for seven matches, cataloged team equipment, created a field marking duty roster and led the offensive midfield in 19 matches.
4. Aviation Club (Deputy Chairperson) - Orchestrated meetings, designed prototype AV24 plane cockpit & trained 300 students in aviation clubs from 4 schools on the same. Featured in WanafunziHewani magazine.

SUMMER ACTIVITIES

1. Designated nanny of my three siblings - every summer/holiday
2. Physics and Mathematics volunteer teacher - July 2022 to present
3. Volunteer at MalishoNiHapa community feeding basket - Every summer from May 2020 to present
4. Tesa Soccer Club Coach - Every summer from May 2021 to present
5. Generation X Solutions LTD intern- May 2022 to July 2022
6. Mwanga Kijijini volunteer-made & distributed homemade candles- Every summer from May 2020- September 2021

SCHOLASTIC DISTINCTIONS

1. Kenya National Examinations Council - 8th student nationally in the Kenya Certificate Of Secondary Education
2. Kenya National Examinations Council - Best in Physics in the Kenya Certificate of Secondary Education
3. Danish Robotics Exchange Program Cohort - Student innovator of the year & overall best team leader
4. The Kenya High School Technovation Team - Most Outstanding Front-end Developer
5. The Kenya High School STEM Club - Communication Optimization Pioneer

NON-SCHOLASTIC DISTINCTIONS

1. Nairobi SUKELLEMO Women's Soccer - Most Valuable Player
2. The Kenya High School's North Club Initiative - Altruist of the Year
3. Nairobi District Public School Orchestra - First Violinist
4. The Kenya High School Entertainment Crew - Operational Excellence Award (Media, Public Address and Leisure Department)
5. Mwanga Kijijini Association - Ultimate Chandler-(Initiated candlelight provision in 33 homes)

MIT ESSAY DRAFTS

Describe the world you come from; community, school etc. How has it shaped your dreams and aspirations?(225 words)

Gloomy. Black. Whatever word reminds you of a darkness so deep that it drenches and drowns you, engulfing you in every way. A magnetic darkness. Candles burning brightly in selfless solidarity, escorting the sun in its retreat behind the Ngong Hills, attempting, but failing, to banish the unwavering darkness. The premature end to a day. A day that, due to the power rationing, would never unlock its full potential. A crippled day.

In this dark town, I battled my first encounter with my nyctophobia. With the candles finally reduced to melted mounds and no similitude of light, I crawled into a corner, my breath hitching in my throat, as the boisterous giggles and guffaws of drunk men in the distance made me inch towards and merge with the cold wall. I had to go through this every night, I hated it.

But, it was in this same dark town that my kerosene lamp illuminated my path to discovery as I delved through a second-hand copy of 'Power to Save the World' by Gwyneth Cravens and began to see the source of the light I so desperately sought. It was in this dark town that I discovered my purpose; A light bringer. And, unconventional though it be, I will do it all with the help of uranium.

Pick what field at MIT appeals to you the most right now. (100 words)

The fictitious Wakanda is all Africa has to show for considerable advancements in the adoption of nuclear power. The real Africa, however, is dealing with inconsistent, sometimes-absent, hydroelectric-power caused by climate change-induced droughts. Consequently, people suffering from nyctophobia, like me, drown in their fears nightly.

At MIT, under the instruction of Prof. Jacopo Buongiorno, I will gather extensive knowledge on Nuclear Batteries and offshore floating nuclear power plants, champion sustainable nuclear power in Africa, and dispel most natural concerns about its safety implications. Nuclear engineering is the key to an Africa where the only thing that's dark is our skin.

What do you do for fun?(225 words)

“Akili ni nywele.” Loosely translated, “The brain is in the hair.”

Ten years ago, you would look at my kinky, disobedient, tumbleweed-like hair and decipher, by all standards that rang true and African, I had absolutely no brain. The knots, wooly balls and stray blanket chunks that had decided to commandeer my ‘brain’ were so vigilant in their occupation that not once did the dreaded enemy, Colonel Comb, successfully make his way through their land. My only consolation was that at certain angles, in certain lighting, with the right mirror, my hair could pass for cotton-candy. Really black cotton-candy.

After months of observing hairdressers, an antidote. Furiously, I tugged at my hair in pursuit of the elusive cornrow. Mistakes were made, tufts were cut, and bald spots created. When my scalp gave up, I moved to my mother’s, my sister’s, my neighbors’, the grass outside; obsessed couldn’t come close to describing me.

Now, in the spirit of full disclosure, I must admit, I’m not about to win any awards for my work. Sometimes my cornrows look less like lines and more like mazes. However, the feel of each strand submitting to my will with each calculated movement births a calm that I spent years chasing after. A calm that I cannot imagine living without. A calm that tames not just hair but my tumultuous soul.

Describe one way in which you have contributed to your community.(225 words)

A procession of children, grime and mud clinging to their chocolate skin, headed behind a derelict brick house. Minutes later, they emerge with battered, brown bags containing bhang. Bhang that they will indulge in and supply, masked by this established anomaly- the Rongai Drug Business.

Poor, rejected by the community, and desperate to survive, these children turned to drug cartels. Musing over how to alleviate the situation, it hit me -soccer. Everybody loves a good old cow-skin chase. Hence, TESA Soccer club was born.

Through TESA, I offer soccer training to 53 children in the drug world. Community tournaments are played and won, prize money distributed and several students re-enrolled in school. But, the club isn't just about soccer. It is a safe space for these children to express all their needs. Several weekends are spent scouring physics textbooks with them, helping raise their grades by a few points, and teaching them how to fashion make-shift light-bulb circuits for their late-night use. And let's not forget the days that all we do is go down to the river and talk about how so and so's father is getting a fresh batch of tomatoes for market day.

TESA Soccer club. It certainly doesn't compare to the likes of Barcelona FC but, it is a home. For anyone that so happens to need one.

Tell us about a significant challenge you've faced and how you overcame it?(225 words)

As they lay on the earthen floor, writhing and groaning simultaneously as if telepathically connected by their mutual pain, one thing and one thing alone became evident. My parents were going to die.

In the thick of the COVID-19 pandemic, my parents contracted the deadly disease. Their wanton coughs punctuated the silence in the home, occasionally interspersed by their equally devastating gasps for breath. The new norm. Every cough and every gasp served as a reminder that there was no medicine, there was no time and there was no hope. I could feel myself fading away as I observed their bodies

morph into the sickness itself and deteriorate from the lack of both food and medicine. Their slow transition from the living to the dead initiated my hunt.

With students out of school, parents sought to ensure that they don't fall behind. Abracadabra; I appear with the magic tuition. Taking on three students for nine hours daily in Physics and Mathematics, I began to make a living for my family. Every single penny I earned was directed to the pharmacist and the spinach vendor on our street. Over-the-counter generic drugs and half a plate of spinach became the fuel to my fire as life returned into my parents' bodies. A small initiative restored some semblance of order in my life.

Significant Challenge Essay - Draft 2 (225 words)

Consent is a fragile thing. My presence was all it took.

The venomous darkness shrouded us, a single candle flame forcefully tethering me back to a body. Not my body. I didn't want it anymore. The flame maneuvered the air with a boastful ease. Artfully emphasizing my own incapacitation. Family is supposed to be harmless. Uncles aren't supposed to pin you down in the dark.

On the 27th of April 2021, I was raped.

Don't look at me like that, he said. You should have stopped me, he said. I, on the other hand, said nothing. To no one. I worked for him, he paid my fees. That was the deal. A deal I had embraced, for my parents' sake, until that night. The following morning, I left. I abandoned a part of me in that house.

On the 27th of August 2021, I attended his funeral. Suicide.

His casket was the same shade of black as that night.

He should have been there. To watch me break my silence and crumble into my father's arms. To see my father crawling to the soccer field with me, sitting patiently as I abandoned his memory in that field. He should have seen me fashion

a new body. One I **wanted**. One that hadn't healed, I admit, but was certainly on the path to redemption.

Gap Year Statement (400 words)

Kenyan high school graduates are required to take a gap year after their KCSE exams.

I spent the first few weeks of my gap year fixing the neighbors' broken home appliances, cleaning out condenser coils, checking Miniature Circuit Breakers and tinkering with saddle valves all for the reward of a generously awarded "Thank You" and on lucky days, a meal afterwards.

From May to July, I interned at GenerationX Solutions LTD, which is a company that researches, presents techniques and deploys projects to combat the lack of adequate electricity in East Africa, and comes up with new methods to avail sustainable energy in East Africa. While at the company, I led a team of seven other interns in in-person data collection, analysis and presentation, for two projects commissioned by the Nuclear Power and Energy Agency and the Kenya Electricity Generating Company. On top of that, I came up with a model for effective public sensitization on new, alternative power generation methods that was adopted in the rollout of the project. I also cataloged the company's project evaluations and gave weekly suggestions on how to improve client satisfaction until mid-July.

Afterwards, I got the opportunity to go through the very rigorous and selective process of joining EducationUSA. More than 2500 of us applied for a spot in the organization; writing essays, taking aptitude tests and attending numerous vetting interviews. Eventually, I was one of the 40 accepted into the program where I have been receiving guidance on how to apply to universities in the USA alongside preparing me for my transition to college.

Whilst I go on with the program, I have been volunteering as a physics and mathematics teacher to the 567 form 4 (grade 12) students at Ongata Ronkai

Secondary School for 21 hours a week, to help mitigate the impact of teacher shortage, which has seen the mean score rise by 40% in both subjects. I also use this opportunity to hold daily evening meetings with the girls where I exhort them to eschew early marriages and pursue their academic goals without letting the patriarchy that prevails in our society relegate them to the housewife's role.

I have also been coaching 53 children in soccer.

OPTIONAL ESSAYS

More about cultural background (150 words)- Optional in the introduction section

Luo. Technically, it's my tribe. In reality, it's simply a creative way of packaging my ancestral CV.

Qualification number one would be my invaluable skill to construct a *simba*. Granted this mud house is ordinarily constructed by young men fresh from the initiation cave, I learnt that if you disrupt their work just enough, by rolling around in their mud, you will soon be apportioned your own section to construct. With enough mud-related disruptions and enough portions assigned, muscle memory kicks in and soon enough you can build a whole house.

Then, there's the fishing. I can navigate a boat through Lake Victoria waters in my sleep. Oar or not. Negotiating my way through the hyacinth with a firmly-established finesse.

And finally, there's the implied guarantee that, like all the women that bore the Luo label before me, I come with an hourglass figure. Spoiler alert, I don't.

Additional information- In the parents' employment section (150 words)

The Kenyan 2007-2008 post election violence claimed many things. Lives, homes, peace and my father's legs. On 15th January three ruffians, intent on eradicating people from my tribe, broke into our shanty in Kibera with makeshift mace clubs. My father, already incapacitated by pneumonia, was the only one in the house with me and my last line of defense. When they barged in, I instinctively fell on my bedridden father's chest, seeking refuge, screaming. This simple act prevented the

men from attacking my father's vital organs whilst he struggled to get up and defend us. However, they turned their sights to his exposed legs and battered them to the point of severance.

He lost his legs and we lost him. Depression and post traumatic stress disorder have confined him to a mat in the house, leaving us at the mercy of my mother's meager earnings.

SUBMITTED

Physics Rec Letter

Talks about:

1. Technovation
2. Danish Exchange Program Cohort
3. Me as a physics student

Music Rec Letter

Talks about:

1. Me as a music Student
2. Football Club
3. How I revived the music department of the school
4. RhythmOfOurSouls music campaign that I started

Counselor Rec Letter

Talks about:

1. Me as a person
2. The North Club Initiative that I started-Students raise funds and items for their needy peers
3. How I transformed the soccer team to be more aware of the needy
4. My Sister's struggle with depression and how it affected me plus how I managed to encourage her

5. My ardent desire to pursue nuclear engineering
6. My work with the school STEM Club that allowed me to get an opportunity to join the Danish Robotics Exchange Program Cohort
7. My work with my team on the Kenya Science and Engineering Fair Project

NOT SUBMITTED BUT ADDED ONTO THE PLATFORM

Corporate Rec Letter

From Generation X Solutions LTD

1. How I was as an employee
2. The work I did in the office
3. The interest I took in their projects with NuPEA (Nuclear Power and Energy Agency) and KenGen (Kenya Electricity Generating Company) and how I contributed to the same.